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Inventor(s)	Iida; Shin

Automatic analyzing device

Abstract

An automatic analyzing device according to an embodiment of the present disclosure includes a rotating table, a plurality of light receiving units, and a light radiating unit. The rotating table includes a plurality of placement parts on which a plurality of reaction cuvettes are placed, respectively. The plurality of light receiving units are provided in correspondence with the plurality of placement parts, respectively. The light radiating unit is configured to change from one of the reaction cuvettes to another on which light is radiated, by changing an emission direction of the light.

Inventors:	Iida; Shin (Ota-ku, JP)
Applicant:	CANON MEDICAL SYSTEMS CORPORATION (Tochigi, JP)
Family ID:	1000008820443
Assignee:	CANON MEDICAL SYSTEMS CORPORATION (Tochigi, JP)
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Primary Examiner: Wallenhorst; Maureen

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-032250, filed on Mar. 2, 2021; the entire contents of which are incorporated herein by reference.

FIELD

(2) The embodiments described herein relate generally to an automatic analyzing device.

BACKGROUND

(3) In an automatic analyzing device for a clinical test, a biological specimen (hereinafter, "specimen") such as blood or urine is mixed with a reagent in predetermined amounts to allow a reaction, so as to obtain the density, an activity value, a time period required by a change, and the like of the substance subject to a measuring process, by measuring the light quantity of transmitted light or scattered light obtained by applying light to the liquid mixture. For example, the automatic analyzing device is configured to allow a photometry unit to radiate the light, so as to perform the measuring process on the liquid being the liquid mixture provided in a reaction cuvette serving as a reaction tube.

(4) The photometry unit includes: a light emitting unit configured to radiate the light toward the reaction cuvette; and a light receiving unit configured to detect the light that has passed through the liquid in the reaction cuvette. In this situation, the light quantity of the radiated light may vary among light emitting units. For this reason, when photometry units are provided in a quantity equal to the number of slots on which reaction cuvettes are placed on a reaction disk, it is necessary to adjust the variation in the light quantity as many times as the number of slots, for the purpose of making uniform the light quantities of the light radiated by the light emitting units of the photometry units.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating an exemplary configuration of an automatic analyzing device according to an embodiment of the present disclosure;

(2) FIG. 2 is a perspective view illustrating an exemplary configuration of an analyzing device of the automatic analyzing device in FIG. 1;

(3) FIG. 3 is a perspective view of a reaction cuvette used in the automatic analyzing device according to the present embodiment;

(4) FIG. 4 is a top view of a reaction disk;

(5) FIG. 5 is a cross-sectional view at A-A in FIG. 4; and

(6) FIG. 6 is a cross-sectional view at A-A in FIG. 4 according to a modification example of the present embodiment.

DETAILED DESCRIPTION

(7) An automatic analyzing device according to an embodiment of the present disclosure includes a rotating table, a plurality of light receiving units, and a light radiating unit. The rotating table includes a plurality of placement parts on which a plurality of reaction cuvettes are placed, respectively. The plurality of light receiving units are provided in correspondence with the plurality of placement parts, respectively. The light radiating unit is configured to change from one of the reaction cuvettes to another on which light is radiated, by changing an emission direction of the light.

(8) Exemplary embodiments of an automatic analyzing device will be explained in detail, with reference to the accompanying drawings. Possible embodiments are not limited to the embodiments described below. Further, in principle, the description of each of the embodiments is similarly applicable to any other embodiment.

(9) FIG. 1 is a block diagram illustrating an exemplary configuration of an automatic analyzing device **100** according to the present embodiment. The automatic analyzing device **100** illustrated in FIG. 1 includes an analyzing device **70**, a driving device **80**, and a processing device **90**.

(10) The analyzing device **70** is configured to generate standard data and tested data, by performing

a measuring process on liquid mixtures in each of which a standard specimen of any of various test items or a tested specimen (a biological specimen such as blood or urine) collected from an examined subject is mixed with a reagent used for an analysis of the test item. The analyzing device **70** includes a plurality of units configured to dispense the specimens, to dispense the reagents, and the like. The driving device **80** is configured to drive the units of the analyzing device **70**. The processing device **90** is configured to bring the units of the analyzing device **70** into operation, by controlling the driving device **80**.

(11) The processing device **90** includes an input device **50**, an output device **40**, a controlling circuitry **30**, and a storage circuit **60**.

(12) The input device **50** includes inputting devices such as a keyboard, a mouse, a button, a touch key panel, and/or the like and is configured to input information for setting analysis parameters of the test items and to input information for setting tested identification information and the test items of the tested specimen.

(13) The output device **40** includes a printer and a display device. The printer is configured to print data generated by the controlling circuitry **30**. The display device is a monitor using a Cathode Ray Tube (CRT), a liquid crystal panel, or the like and is configured to display data generated by the controlling circuitry **30**.

(14) The storage circuit **60** is, for example, a semiconductor memory element such as a Random Access Memory (RAM) or a flash memory, or a storage device such as a hard disk or an optical disk.

(15) The controlling circuitry **30** is configured to control the entirety of a system. For example, as illustrated in FIG. **1**, the controlling circuitry **30** is configured to execute a data processing function **31** and a controlling function **32**. The controlling function **32** is configured to bring the units of the analyzing device **70** into operation, by controlling the driving device **80**. In this situation, the controlling function **32** is an example of a controlling unit. The data processing function **31** is configured to generate calibration data and analysis data of each of the test items, by processing the standard data and the tested data generated by the analyzing device **70**.

(16) For example, the standard data generated by the analyzing device **70** denotes data (a calibration curve or a standard curve) for determining an amount or the density of a substance. The tested data generated by the analyzing device **70** denotes data of a result from the measuring process performed on the tested specimen. Further, the calibration data output from the controlling circuitry **30** denotes data indicating a result of the measuring process such as the amount or the density of the substance derived from the tested data and the standard data. The analysis data output from the controlling circuitry **30** denotes data indicating a result of determining positive or negative. In other words, the calibration data is data for deriving the analysis data indicating the result of determining positive or negative.

(17) In this situation, for example, processing functions executed by constituent elements of the controlling circuitry **30** are recorded in the storage circuit **60** in the form of computer-executable programs. The controlling circuitry **30** is a processor configured to realize the functions corresponding to the programs by reading and executing the programs from the storage circuit **60**. In other words, the controlling circuitry **30** that has read the programs has the functions illustrated within the controlling circuitry **30** in FIG. **1**.

(18) Although the example is explained with reference to FIG. **1** in which the single controlling circuit (the controlling circuitry **30**) realizes the processing functions described below, it is also acceptable to structure a processing circuit by combining together a plurality of independent processors, so that the functions are realized as a result of the processors executing the programs.

(19) The term “processor” used in the above explanations denotes, for example, a Central Processing Unit (CPU), a Graphics Processing Unit (GPU), or a circuit such as an Application Specific Integrated Circuit (ASIC) or a programmable logic device (e.g., a Simple Programmable Logic Device [SPLD], a Complex Programmable Logic Device [CPLD], or a Field Programmable

Gate Array [FPGA]). When the processor is a CPU for example, the processor realizes the functions by reading and executing the programs saved in the storage circuit **60**. In contrast, when the processor is an ASIC, for example, instead of the programs being saved in the storage circuit **60**, the programs are directly incorporated in the circuit of the processor. Further, the processors of the present embodiments do not each necessarily have to be structured as a single circuit. It is also acceptable to structure one processor by combining together a plurality of independent circuits, so as to realize the functions thereof. Further, it is also acceptable to integrate two or more of the constituent elements in FIG. **1** into a single processor, so as to realize the functions thereof.

(20) FIG. **2** is a perspective view illustrating an exemplary configuration of the analyzing device **70** of the automatic analyzing device **100** in FIG. **1**.

(21) For example, the analyzing device **70** is used for a blood coagulation test. The analyzing device **70** includes a reaction disk **10**, a constant temperature part **11**, a rack sampler **12**, and a reagent storage **14**.

(22) The reaction disk **10** includes a plurality of slots **10a** on which reaction cuvettes **1** serving as reaction tubes are placed. Further, the reaction disk **10** rotates on a rotation axis R extending parallel to the vertical direction. The plurality of slots **10a** are provided while being arranged along a circumferential direction centered on the rotation axis R of the reaction disk **10**. Each of the slots **10a** is a groove recessed downward from the top face of the reaction disk **10**. As the reaction cuvettes **1** are placed on the slots **10a**, the reaction disk **10** holds the plurality of reaction cuvettes **1** that are arranged in an annular formation. As a result of driving of the driving device **80** so as to repeatedly alternate between rotating and stopping at predetermined time intervals while the reaction cuvettes **1** are held on the reaction disk **10**, the reaction disk **10** transports the reaction cuvettes **1** held thereby. In this situation, the reaction disk **10** is an example of the rotating table, whereas the slots **10a** are an example of the placement parts.

(23) The constant temperature part **11** has stored therein a heat medium set to a predetermined temperature and is configured to increase the temperature of a liquid mixture contained in the reaction cuvettes **1** by immersing the reaction cuvettes **1** in the heat medium being stored.

(24) The rack sampler **12** is configured to movably support specimen racks **13** each capable of holding a plurality of specimen containers that contain specimens for which a measuring process was requested. FIG. **2** illustrates the specimen racks **13** each capable of holding five specimen containers arranged in a row.

(25) The rack sampler **12** is provided with a first region where each of the specimen racks **13** is transported from an input position in which the specimen rack **13** is input, to a collection position in which the specimen rack **13** on which the measuring process has been completed is collected. In the first region, the plurality of specimen racks **13** arranged in the longitudinal direction are driven by the driving device **80** so as to be moved in a direction D1.

(26) Further, the rack sampler **12** is provided with a second region where each of the specimen racks **13** is drawn in from the first region, so that the specimen containers held by the specimen rack **13** are moved to a predetermined sample suction position. The sample suction position is provided, for example, in the position where a rotation track of a sample dispensing probe (explained later) intersects a moving track of an opening part of each of the specimen containers supported by the rack sampler **12** and held by the specimen rack **13**. In the second region, each of the specimen racks **13** that has been transported is driven by the driving device **80** so as to be moved in a direction D2.

(27) Further, the rack sampler **12** is provided with a third region where each of the specimen racks **13** holding the specimen containers from which the specimens were aspirated out is returned to the first region. In the third region, each of the specimen racks **13** is driven by the driving device **80** so as to be moved in a direction D3.

(28) The reagent storage **14** is configured to keep cool a plurality of reagent containers **15** each storing therein a reagent or the like used for the various test items performed on the standard liquid

and the specimens. The reagent storage **14** includes a reagent storage cover **14a** and a reagent rack **14b**.

(29) The reagent storage **14** is covered by the reagent storage cover **14a** that is detachably attached. The reagent rack **14b** is provided so as to be rotatable. The reagent rack **14b** is configured to hold the plurality of reagent containers **15** placed in an annular formation. For example, the reagent containers **15** are each a columnar container made of glass. More specifically, the reagent containers **15** are each a container made of glass having the shape of a circular column or the shape of a polygonal prism that can be housed in a predetermined circular cylinder.

(30) The analyzing device **70** further includes a sample dispensing arm **16**, a sample dispensing probe **17**, a reagent dispensing arm **18**, and a reagent dispensing probe **19**.

(31) The sample dispensing arm **16** is provided between the reaction disk **10** and the rack sampler **12**. The sample dispensing arm **16** is provided so as to be movable up and down along the vertical directions and so as to be rotatable in the horizontal direction. The sample dispensing arm **16** holds, on one end thereof, the sample dispensing probe **17**.

(32) The sample dispensing probe **17** is configured to turn in an arc form, in conjunction with the rotation of the sample dispensing arm **16**. On the rotation track, the sample suction position is provided in which the specimen is aspirated out of each of the specimen containers held by the specimen racks **13** on the rack sampler **12**. Further, on the rotation track of the sample dispensing probe **17**, a sample dispense position is provided in which the specimen aspirated by the sample dispensing probe **17** is dispensed into each of the reaction cuvettes **1**. The sample dispense position corresponds, for example, to an intersection of the rotation track of the sample dispensing probe **17** and a moving track of each of the reaction cuvettes **1** held by the reaction disk **10**.

(33) The sample dispensing probe **17** is driven by the driving device **80** so as to be moved in the up-and-down directions in the sample suction position or in the sample dispense position. Further, under control of the controlling circuitry **30**, the sample dispensing probe **17** is configured to aspirate the specimen out of one of the specimen containers stopped in the sample suction position. Further, under control of the controlling circuitry **30**, the sample dispensing probe **17** is configured to dispense the aspirated specimen into one of the reaction cuvettes **1** stopped in the sample dispense position.

(34) The reagent dispensing arm **18** is provided between the reaction disk **10** and the reagent storage **14**. The reagent dispensing arm **18** is provided so as to be movable up and down along the vertical directions and so as to be rotatable in the horizontal direction. The reagent dispensing arm **18** holds, on one end thereof, the reagent dispensing probe **19**.

(35) The reagent dispensing probe **19** is configured to turn in an arc form, in conjunction with the rotation of the reagent dispensing arm **18**. On the rotation track, a reagent suction position **H1** is provided. The reagent suction position **H1** is provided in the position where the rotation track of the reagent dispensing probe **19** intersects a moving track of an opening part of each of the reagent containers **15** placed on the reagent rack **14b** in the annular formation. Further, on the rotation track of the reagent dispensing probe **19**, a reagent dispense position is provided in which the reagent aspirated by the reagent dispensing probe **19** is dispensed into each of the reaction cuvettes **1**. For example, the reagent dispense position corresponds, for example, to an intersection of the rotation track of the reagent dispensing probe **19** and the moving track of each of the reaction cuvettes **1** held by the reaction disk **10**.

(36) The reagent dispensing probe **19** is driven by the driving device **80** so as to be moved in the up-and-down directions in the reagent suction position **H1** or in the reagent dispense position on the rotation track. Further, under control of the controlling circuitry **30**, the reagent dispensing probe **19** is configured to aspirate the reagent out of one of the reagent containers stopped in the reagent suction position **H1**. Further, under control of the controlling circuitry **30**, the reagent dispensing probe **19** is configured to dispense the aspirated reagent into one of the reaction cuvettes **1** stopped in the reagent dispense position.

(37) The analyzing device **70** further includes: a cleaning tank used for cleaning the sample dispensing probe **17** every time the dispensing of the specimen is finished; another cleaning tank used for cleaning the reagent dispensing probe **19** every time the dispensing of the reagent is finished; an agitator configured to agitate the liquid mixture in the reaction cuvettes **1**; and another cleaning tank used for cleaning the agitator every time the agitation is finished.

(38) The analyzing device **70** further includes a photometry unit **20** and a cleaning unit. The photometry unit **20** is configured to perform a measuring process on the liquid mixtures in the reaction cuvettes **1** by radiating light thereon. The cleaning unit is configured to clean the inside of the reaction cuvettes **1** on which the photometry unit **20** has finished performing the measuring process. The photometry unit **20** will be explained later.

(39) The driving device **80** is configured to drive the units of the analyzing device **70**.

(40) The driving device **80** includes a mechanism configured to drive the rack sampler **12** of the analyzing device **70** and is configured to move the specimen racks **13**. Further, the driving device **80** includes a mechanism configured to drive the reagent rack **14b** of the reagent storage **14** and is configured to rotate the reagent containers **15**. Further, the driving device **80** includes a mechanism configured to drive the reaction disk **10** and is configured to rotate the reaction cuvettes **1**.

(41) Further, the driving device **80** includes a mechanism configured to rotate and move up and down the sample dispensing arm **16** and is configured to move the sample dispensing arm **16** between the specimen containers and the reaction cuvettes **1**. Further, the driving device **80** includes a mechanism configured to drive the sample dispensing probe **17** and is configured to allow the sample dispensing probe **17** to dispense the specimen, by causing the sample dispensing probe **17** to aspirate the specimen out of each of the specimen containers and to dispense the specimen into each of reaction cuvettes **1**.

(42) Further, the driving device **80** includes a mechanism configured to rotate and move up and down the reagent dispensing arm **18** and is configured to move the reagent dispensing arm **18** between the reagent containers **15** and the reaction cuvettes **1**. Further, the driving device **80** includes a mechanism configured to drive the reagent dispensing probe **19** and is configured to allow the reagent dispensing probe **19** to dispense the reagent, by causing the reagent dispensing probe **19** to aspirate the reagent out of each of the reagent containers **15** and to dispense the reagent into each of the reaction cuvettes **1**.

(43) Next, the reaction cuvettes **1** will be explained, with reference to FIG. 3. FIG. 3 is a perspective view of one of the reaction cuvettes **1** used in the automatic analyzing device **100** according to the present embodiment.

(44) As illustrated in FIG. 3, the reaction cuvette **1** has a trunk part **1a**, a crown part **1b**, and a flange **1c**.

(45) The trunk part **1a** is a member having the shape of a bottomed circular cylinder of which the top face is open. The trunk part **1a** has formed therein a plurality of photometry sections **1d**. For example, the trunk part **1a** has formed therein four photometry sections **1d**. Each of the four photometry sections **1d** has a planar shape, for example.

(46) The flange **1c** is a ring-shaped member. The outside diameter of the flange **1c** is larger than the outside diameter of the trunk part **1a**. Further, the flange **1c** is provided on the opening (**1e**) side of the trunk part **1a**. The specimen or the reagent is dispensed into the trunk part **1a** from the opening (**1e**) side.

(47) The crown part **1b** is a hollow member having the shape of a rectangular cylinder and has four planar faces. The four planar faces are provided so as to stand on the top face of the flange **1c**. In a top view, the shape of the crown part **1b** is substantially rectangular. The planar faces of the crown part **1b** are provided so as to correspond to the photometry sections **1d** of the trunk part **1a**. The light from the photometry unit **20** illustrated in FIG. 2 is radiated onto the photometry sections **1d**.

(48) As explained above, the photometry unit **20** is configured to perform the measuring process on the liquid in the reaction cuvettes **1** by radiating the light thereon. More specifically, the

photometry unit **20** is configured to measure optical property values in the liquid mixture of the specimen and the reagent dispensed in the reaction cuvette **1**. Further, if photometry units **20** each included a light emitting unit configured to radiate the light toward a reaction cuvette **1** and a light receiving unit configured to detect transmitted light or scattered light obtained by applying the light to the liquid mixture in the reaction cuvette **1**, while the photometry units **20** were provided in a quantity equal to the number of slots **10a** on which the reaction cuvettes **1** are placed on the reaction disk **10**, it would be necessary to adjust the variation in the light quantity among the light emitting units, as many times as the number of slots **10a**, for the purpose of making uniform the light quantities of the light radiated by the light emitting units. Further, to adjust the variation in the light quantity among the light emitting units as many times as the number of slots **10a**, adjusting the variation in the light quantity would require time and effort as well as labor costs.

(49) Further, due to aging deterioration of the light emitting units, it might become necessary in some situations to provide a compensation circuit that compensates a decrease in the light quantity of the deteriorating light emitting units. In those situations, when the light quantity decreases due to the aging deterioration of the light emitting units even with the use of the compensation circuit, there would be a possibility that as many slots **10a** as the number of deteriorating light emitting units might become unusable.

(50) To cope with this situation, for the purpose of making it possible to perform stable analyses, the automatic analyzing device **100** according to the present embodiment includes the reaction disk **10** serving as the rotating table, the plurality of light receiving units, and a light radiating unit. The reaction disk **10** has the plurality of slots **10a** on which the plurality of reaction cuvettes **1** are placed, respectively. The plurality of light receiving units are provided in correspondence with the plurality of slots **10a**, respectively. The light radiating unit is configured to change from one of the reaction cuvettes **1** to another on which the light is radiated, by changing the emission direction of the light.

(51) FIG. **4** is a top view of the reaction disk **10**. FIG. **4** illustrates only some of the plurality of slots **10a** provided on the reaction disk **10** and some the plurality of reaction cuvettes **1** held in the plurality of slots **10a**. The others are omitted because the configurations thereof are the same.

(52) The photometry unit **20** includes a light radiating unit **21** and a plurality of light receiving units **22**. The single light radiating unit **21** is provided in a position corresponding to the rotation axis R of the reaction disk **10**. In other words, the light radiating unit **21** is provided at the center of the reaction disk **10**. In the reaction disk **10**, the light radiating unit **21** is configured to emit light (the light L in FIG. **2**) toward each of the plurality of reaction cuvettes **1**, by changing the emission direction of the light from the center of the reaction disk **10**.

(53) The light receiving units **22** are provided in a quantity equal to the number of slots **10a** of the reaction disk **10**. In other words, the light receiving units **22** are provided in the quantity equal to the number of reaction cuvettes **1** held in the slots **10a** of the reaction disk **10**. When detecting the light transmitted through the liquid mixture in a reaction cuvette **1**, the light receiving unit **22** is positioned on the outside (the outer circumferential side) of the slot **10a** in terms of the radial direction. In another example, when detecting the light scattered by particles in the liquid mixture in a reaction cuvette **1**, the light receiving unit **22** is positioned on the outside of the slot **10a** in terms of the circumferential direction.

(54) FIG. **5** is a cross-sectional view at A-A in FIG. **4**. The light radiating unit **21** includes a light emitting unit **21a**, a mirror **21b**, and a motor **21c**. In the present embodiment, a mechanism having a single light source is realized by making the light radiating unit **21** a rotating type.

(55) The light emitting unit **21a** is provided in a position in a bottom face part of the reaction disk **10** that corresponds to the rotation axis R of the reaction disk **10**. The light emitting unit **21a** is, for example, a Light Emitting Diode (LED) or a laser light source and is configured to radiate the light L.

(56) The motor **21c** includes a rotation axis and a main body part configured to rotate the rotation

axis. In a top face part of the reaction disk **10**, the motor **21c** is provided in a position corresponding to the rotation axis R of the reaction disk **10**. The rotation axis of the motor **21c** rotates on the same axis as the rotation axis R of the reaction disk **10**. As a result of being driven by the driving device **80**, the main body part of the motor **21c** allows the rotation axis of the motor **21c** to rotate. The motor **21c** or the rotation axis of the motor **21c** is an example of the rotating unit.

(57) The mirror **21b** is provided at a tip end of the rotation axis of the motor **21c**. For example, the mirror **21b** is configured to emit the light L toward a reaction cuvette **1**, by reflecting the light L radiated from the light emitting unit **21a** at a 90-degree angle. The mirror **21b** is configured to rotate in conjunction with the rotation of the motor **21c**. The light L emitted from the light emitting unit **21a** is reflected by the mirror **21b** and subsequently travels toward the slot **10a** and the light receiving unit **22** along the radial direction of the reaction disk **10**.

(58) As a result, the mirror **21b** emits the light L toward each of the photometry sections **1d** of the plurality of reaction cuvettes **1**, by reflecting the light L radiated from the light emitting unit **21a** and changing the emission direction of the light L in conjunction with the rotation of the motor **21c**. For example, as for the light emitting unit **21a**, the mirror **21b** rotates due to the rotation of the motor **21c**, so that the light L is emitted onto each of the plurality of reaction cuvettes **1** at predetermined time intervals. The predetermined time intervals are, for example, intervals of 0.1 seconds.

(59) While the reaction cuvettes **1** injected with the liquid mixture are placed on the slots **10a**, the light L is emitted from the light radiating unit **21** onto the photometry section **1d** of a reaction cuvette **1**. In this situation, the light receiving unit **22** is configured to detect transmitted light or scattered light obtained by applying the light to the liquid mixture in the reaction cuvette **1**. The light receiving unit **22** is configured to sample the detected light at predetermined time intervals (e.g., intervals of 0.1 seconds). Further, the light receiving unit **22** is configured to measure the light quantity, the intensity, and/or the like of the detected light and to output a measurement result to the controlling circuitry **30** as tested data.

(60) Although the light emitting unit **21a** is provided in the reaction disk **10** in the example in FIG. 5, the light emitting unit **21a** may be provided on the outside of the reaction disk **10**, for example. For instance, the reaction disk **10** may be provided with a through hole formed in a bottom face part thereof, so that the light emitting unit **21a** radiates the light L from the outside of the reaction disk **10** via the through hole.

(61) Further, although the motor **21c** is provided in the reaction disk **10** in the example in FIG. 5, the main body of the motor **21c** may be provided on the outside of the reaction disk **10**, for example. For instance, the reaction disk **10** may be provided with a through hole formed in the top face part thereof, so that the rotation axis of the motor **21c** extends therethrough.

(62) Further, although in the example in FIG. 5 the light emitting unit **21a** and the motor **21c** are provided on the bottom face side and the top face side of the reaction disk **10**, respectively, the opposite positional arrangement may be used. For example, the light emitting unit **21a** and the motor **21c** may be provided on the top face side and the bottom face side of the reaction disk **10**, respectively.

(63) As explained above, in the present embodiment, the mechanism having the single light source is realized by making the light radiating unit **21** the rotating type. In other words, in the present embodiment, the reaction disk **10** includes the plurality of slots **10a** on which the plurality of reaction cuvettes **1** are placed, respectively. The plurality of light receiving units **22** are provided in correspondence with the plurality of slots **10a**, respectively. The light radiating unit **21** is configured to change from one of the reaction cuvettes **1** to another on which the light L is radiated, by changing the emission direction of the light. As explained herein, according to the present embodiment, the mechanism having the single light source is realized by making the light radiating unit **21** the rotating type. As a result, the light quantity of the light emitting unit is constant, and there is no variation in the light quantity. It is therefore possible to perform stable analyses.

(64) Further, in the present embodiment, the mechanism having the single light source is realized by making the light radiating unit **21** the rotating type. Accordingly, there is no variation in adjustment range where the light quantity of the light L radiated by the light radiating unit **21** is adjusted. Optimization of the adjustment range is thus realized. In other words, in the present embodiment, because adjusting the light quantity of the single light radiating unit **21** alone is sufficient, it is possible to set a wide dynamic range.

(65) Further, in the present embodiment, because adjusting the light quantity of the single light radiating unit **21** alone is sufficient, it is possible to reduce the time and effort and the labor costs required by the adjustment of the variation in the light quantity.

(66) Further, in the present embodiment, the mechanism having the single light source is realized by making the light radiating unit **21** the rotating type. Accordingly, when the single light emitting unit **21a** has aged and become deteriorated, the light emitting unit **21a** can be replaced. As explained herein, in the present embodiment, robustness is expected to improve, in comparison to the situation where as many slots **10a** as the number of aged and deteriorated light emitting units would become unusable.

(67) Other Embodiments

(68) Some embodiments have thus been explained. It is, however, possible to carry out the present disclosure in various different forms other than those described in the above embodiments.

(69) A Modification Example

(70) In the present embodiment, for example, when the light emitting unit **21a** of the light radiating unit **21** is an LED, a problem may arise where the light quantity of the light L radiated from the light emitting unit **21a** attenuates, due to widening of the width of an optical path (the optical path illustrated in FIG. 5) or the like, the optical path extending from the light emitting unit **21a** via the mirror **21b** to each of the light receiving units **22**. In that situation, in a modification example of the present embodiment, light guiding ducts **23** each having a reflection surface on the inside thereof may be provided in the reaction disk **10**, as illustrated in FIG. 6.

(71) The light guiding ducts **23** are provided in a quantity equal to the number of slots **10a** of the reaction disk **10**. In other words, the light guiding ducts **23** are provided in the quantity equal to the number of reaction cuvettes **1** held in the slots **10a** of the reaction disk **10**. For example, the light guiding ducts **23** may be optical fibers. For example, the light guiding ducts **23** include an optical fiber **23a** and a plurality of optical fibers **23b**. In the reaction disk **10**, the optical fiber **23a** is provided between the light emitting unit **21a** and the mirror **21b**. Each of the plurality of optical fibers **23b** is provided in a different one of the spaces between the mirror **21b** and the plurality of light receiving units **22**.

(72) In the present modification example of the embodiment, as a result of providing the light guiding ducts **23** in the reaction disk **10**, it is possible to reduce the attenuation of the light quantity of the light L radiated from the light emitting unit **21a**, so that the light L radiated from the light emitting unit **21a** is guided from the light emitting unit **21a** via the mirror **21b** to the light receiving units **22**. As explained herein, in the modification example, because the mechanism having the single light source is realized by making the light radiating unit **21** the rotating type, the light quantity of the light emitting unit becomes constant. Further, as a result of providing the light guiding ducts **23** in the reaction disk **10**, the attenuation of the light quantity of the light L radiated from the light emitting unit **21a** is reduced. It is therefore possible to perform stable analyses.

(73) According to at least one aspect of the embodiments described above, it is possible to perform stable analyses.

(74) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their

equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. An automatic analyzing device, comprising: a rotating table including a plurality of placement parts on which a plurality of reaction cuvettes are placed, respectively, a plurality of light receiving units provided in correspondence with the plurality of placement parts, respectively; a light radiating unit configured to change from one of the reaction cuvettes to another on which light is radiated, the light radiating unit including a rotating unit configured to rotate on a same axis as a rotation axis of the rotating table: a light emitting unit configured to radiate light from a center of the rotating table, and a mirror connected to the rotating unit and configured to change an emission direction of the light from the center of the rotating table by reflecting the light radiated from the light emitting unit, in conjunction with the rotation of the rotating unit; a first light guiding duct located between the light emitting unit and the mirror, wherein the first light guiding duct is configured to conduct light radiated from the light emitting unit to the mirror; and a plurality of second light guiding ducts (1) provided in correspondence with the plurality of placement parts, which are provided along a circumferential direction centered on the rotation axis of the rotating table, respectively, and (2) configured to guide the light received at the mirror from the light emitting unit via the first light guiding duct, to the plurality of light receiving units, wherein the plurality of second light guiding ducts are located between the mirror and each of the plurality of light receiving units.
